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>Key statistical terminologies

Variable

- Before we can select which statistical method is most appropriate, here are some keywords we need to understand.
- **Variable** - characteristic which varies between independent subjects.
- They are things that we **measure, control, or manipulate in research**.
- Several variables exist in a dataset.
- They differ in many respects, most notably in the **role** they are given in our research and in the **type of measures** that can be applied to them.

Dependent and independent variables

- **Independent** variables are **manipulated** in a scientific experiment to test the effects of the dependent variable.
- The **dependent** variables are only **measured** or **tested**.
- Independent variable is ***changed by the researcher*** intentionally to see how it affects the dependent variable.
- Dependent variable is what the research measures.
- The dependent variable is "dependent" on the independent variable.
- As the independent variable is changed, the effects on the dependent variable is observed and recorded.

Dependent and independent variables

- For example, in an experiment, males are compared to females regarding white cell count (WCC).
- State H0 and H1????
- Gender could be called the **independent** variable and WCC the **dependent** variable.
- We collect data on WCC, and we examine and look at changes in the gender variable to see how it affects WCC.
- Therefore gender is the independent variable.

Relations between variables

- Regardless of variable type eg nominal, ordinal, interval, ratio, two or more variables are related if their ***values are distributed in a consistent manner***.
- Variables are related if their values correspond to each other either ***positively or negatively.***
- E.g. Gender and WCC would be considered related if most males had high WCC and most females low WCC.
- Height is related to weight because, typically, tall individuals are heavier than short ones.
- IQ is related to the Number of Errors in a test if people with higher IQ's make fewer errors.
- Finding a relation between variables – basis of correlational research.
- Statistics helps us evaluate variable correlations & inter-variable relationships

Normal distribution

- Lots of tests based on normal distribution of variables.
- If variables analysed are normally distributed in the population, then they meet the "**normality assumption**."
- Many variables are normally distributed, which is why the normal distribution represents a "**general feature**" of reality.
- Problem may occur when we use a ***normal distribution-based test*** to analyse data from variables that are ***not normally distributed***.
- Then we have two general choices.

Normal distribution

- **First**, use some alternative "nonparametric" test or "distribution-free test“.
- Alternatively (in many cases) we can still use the normal distribution-based test if we only make sure that the size of our samples is **large enough**.
- This is based on important principle - as sample size increases, the shape of the sampling distribution (i.e. distribution of a statistic from the sample) approaches normal shape, even if the distribution of the variable in question is not normal.
- This principle is called the **central limit theorem**.

Parametric test

- **Parametric test** - there are assumptions for parametric tests including the assumption that **continuous (scaled) dependent variables are normally distributed.**

- There are specific parametric tests for this within R, but plotting a histogram is also a good guide.

- As long as the histogram of the dependent variable peaks in the middle and is roughly symmetrical about the mean, we can assume the data is normally distributed.

How do I choose a statistical method?

- Before we can select which statistical method is most appropriate, here's the main questions we need to think about…

- What is your research question?
- Which variables will help you answer your research question?
  - Which variable is the **dependent** variable?
- What **type** of variables are they?
- **Which** statistical test is most appropriate?
- Should a **parametric** or **non-parametric** test be used?

Statistical assumptions

- **Independence of observations** (no autocorrelation): The observations/variables you include in your test are not related (for example, multiple measurements of a single test subject are not independent, while measurements of multiple different test subjects are independent).

- **Homogeneity of variance**: the variance within each group being compared is similar among all groups. If one group has much more variation than others, it will limit the test's effectiveness.

- **Normality of data**: the data follows a normal distribution (a bell curve). This assumption applies only to quantitative data.